

Sang Hyup “Shawn” Lee

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EDUCATION

Johns Hopkins University

May 2026

Master of Science in Computer Science (Combined B.S./M.S.), GPA: 3.87/4.0

Baltimore, Maryland

- **Teaching Assistant:** Object Oriented Software Engineering (Fall 2025, Spring 2026)
- **Relevant Coursework:** Data Structures, Algorithms, Operating Systems, Computer Networks, Databases, Object Oriented Software Engineering, Computer Graphics, Data Science, Software System Design, Augmented Reality, Deep Learning, Artificial Intelligence, Computer Vision

TECHNICAL SKILLS

Languages: Java, Python, C/C++, C#, Rust, JavaScript, Dart, Go, SQL

Frameworks & Libraries: PyTorch, LangChain, React, Django, FastAPI, Node.js, Flutter

Tools & Databases: Docker, GCP, Redis, Firebase, Supabase, Render, MySQL, PostgreSQL, MongoDB

EXPERIENCE

Oracle

May 2025 – August 2025

Software Developer Intern

Redwood City, California

- Built an AI-powered code search agent using Ollama local LLM, enabling natural language queries across 300+ component tags and significantly improving developer search efficiency.
- Developed chat interface using Oracle Visual Builder (JavaScript) with Python/Go backend services, enabling seamless natural language interactions with the code search system.
- Increased query accuracy from 30% to 90% by implementing automated prompting system with DSPy and Instructor, replacing basic prompt engineering approaches.

Quest2Learn

September 2024 – May 2025

Web Developer

Baltimore, Maryland

- Developed virtual labs using React and React 3 Fiber, integrating generative AI for adaptive learning experiences with 400+ students at Johns Hopkins University.
- Built 3 interactive chemistry lab simulations with reusable Three.js components, reducing development time for new lab modules by 500+ lines of code.

JHU Delineo Disease Modeling

September 2024 – December 2024

Data Science Researcher

Baltimore, Maryland

- Leveraged SafeGraph data with NumPy and Pandas to model disease spread across 3,000+ people, analyzing movement patterns across 1,700 households and 2,200 locations.
- Implemented Monte Carlo simulations with probabilistic movement models and dwell time CDFs, processing 30-day mobility patterns with optimized NumPy vectorization reducing runtime by 60%.

Samsung SDS

June 2024 – August 2024

Software Engineering Intern

Seoul, South Korea

- Improved AI chatbot accuracy by 20% using RAG architecture, Few-Shot prompting, and Chain of Thought techniques with Python and LangChain.
- Built vector-based retrieval system using Hugging Face embeddings and cosine similarity search to enhance query understanding and response relevance.

PROJECTS

NooriAI | nooriai.com | *MarianNMT, OpenAI Whisper, VOSK*

February 2025 – Present

- Developed HIPAA-compliant neural machine translation system for translating complex medical terminology to assist Limited English Proficient patients in healthcare settings.

Summit | summit-jobs.com | *React, Supabase, Render, GCP, Redis, Celery*

January 2025 – May 2025

- Built scalable email processing system using Celery, Redis, and Django REST Framework to extract job-related data from Gmail with 92.6% precision, automating job tracking directly from users' inboxes.
- Integrated DistilBERT-based email classifier to filter irrelevant emails with 96.5% accuracy, reducing OpenAI token usage and response latency by 40%.
- Created semantic search chatbot over 15K+ job postings using OpenAI embeddings and Supabase pgvector, providing users with personalized job recommendations in real-time.